

ADR-0001: SaaS Foundation Separate from Business Modules

Project: SMEsPlus Enterprise Suite

Status: Approved for Functional Design

Decision Owner: Project Owner

AI Role: ChatGPT / Virtual Enterprise Solution Architect

Repository Workspace: 99_SMEsPlus_Enterprise_Suite

1. Decision

SMEsPlus Enterprise Suite shall separate **Enterprise SaaS Foundation** from **Business ERP Modules**.

Foundation will provide shared platform capabilities used by every module.

Business modules will focus only on domain-specific ERP functionality.

2. Architecture Structure

SMEsPlus Enterprise Suite will be structured as:

Enterprise Foundation

→ SaaS Platform

→ Security & Identity

→ Subscription & Package Control

→ Configuration Center

→ AI Collaboration Layer

→ Integration Layer

→ Audit & Governance

→ ERP Business Modules

3. Foundation Domains

FD-001 Tenant Management

Supports multi-tenant SaaS operation.

Core Functions:

- Create tenant
- Maintain tenant profile
- Tenant status control
- Company and branch mapping
- Tenant lifecycle: Trial, Active, Suspended, Cancelled, Archived

FD-002 Identity and Access Management

Controls users, roles, and permissions.

Core Functions:

- User registration / invitation
- Login and session control
- Role-based access control
- Module permission
- Company and branch permission
- Auditor read-only access

FD-003 Subscription and Package Management

Controls commercial SaaS entitlement.

Core Functions:

- Plan management
- Module entitlement
- User limit
- Storage limit
- Transaction limit
- Billing cycle
- Trial and renewal
- Upgrade / downgrade

FD-004 Module Activation

Controls which ERP modules are available per tenant.

Core Functions:

- Enable module
- Disable module
- Trial module
- Dependency validation
- Package-based feature control

Example Dependency:

- Sales Invoice requires Accounting
- Delivery requires Inventory
- Purchase Receipt requires Inventory
- Vendor Bill requires Accounting

FD-005 Configuration Center

Central place for tenant-specific configuration.

Core Functions:

- Company settings
- Tax settings
- Currency settings
- Number sequence
- Approval matrix
- Workflow rules
- Notification settings
- Integration settings

FD-006 AI Assistance Layer

Supports AI-assisted ERP operation.

Core Functions:

- Draft functional document
- Summarize transaction
- Generate Jira task
- Generate UAT scenario
- Detect missing evidence
- Prepare Claude handoff
- Generate executive summary

FD-007 Integration Platform

Supports external system integration.

Core Functions:

- REST API
- Webhook
- Event bus
- API token
- Integration log
- Retry and failure handling

FD-008 Audit and Governance

Supports traceability and compliance.

Core Functions:

- Audit log
- Status change history
- Approval history
- Evidence reference
- User activity log
- Review gate status

4. Business Modules

Business modules must use Foundation services instead of creating their own duplicated logic.

Modules:

- CRM Lead Management
- Sales Order to Cash
- Purchase Procure to Pay
- Inventory and Warehouse
- Manufacturing MRP
- Accounting and Finance
- Human Resources
- Project and Helpdesk
- Documents and Approval
- Executive Dashboard

5. Design Rule

Every module must reference Foundation capabilities for:

- Tenant
- Company
- User
- Role
- Permission
- Subscription
- Module activation
- Approval
- Audit
- Notification
- Configuration
- AI task handoff

6. Functional Design Impact

Before designing each module, the team must confirm:

1. Which tenant uses the module
2. Which package enables the module
3. Which roles can access it
4. Which approval rules apply
5. Which master data are required
6. Which configuration values are tenant-specific
7. Which audit events must be recorded
8. Which Jira and Claude handoff records are required

7. Gate Rule

No module can proceed to Claude implementation unless:

- SaaS entitlement is defined
- Role permission is defined
- Functional requirement is documented
- Business rule is documented
- Screen/API/Data object is identified
- UAT case exists
- Evidence gap is cleared or accepted with control

8. Approved Direction

Proceed with SaaS Foundation functional design first.

After Foundation is defined, continue with modules in this order:

1. Purchase
2. Sales
3. Inventory
4. Accounting
5. Documents and Approval
6. Executive Dashboard
7. CRM
8. Manufacturing
9. Project / Helpdesk
10. Human Resources